4.7. POLAR CURVES

(1) Angle between radius vector and tangent. *If ϕ be the angle between the radius vector and the tangent at any point of the curve* $r = f(\theta)$, $\tan \theta = r \dfrac{d\theta}{dr}$.

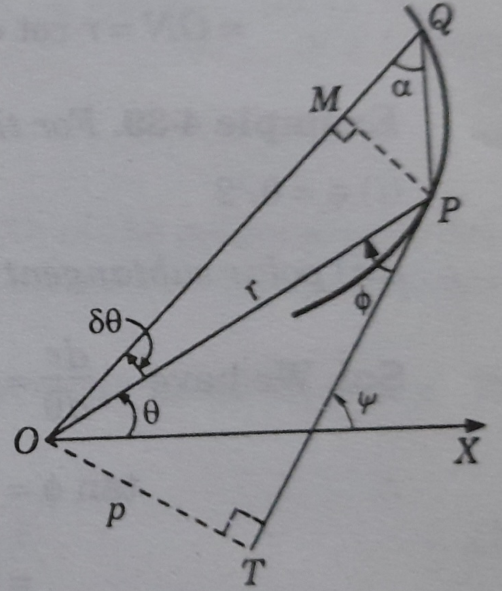


Fig. 4.5.

Let $P(r, \theta)$ and $Q(r + \delta r, \theta + \delta\theta)$ be two neighbouring points on the curve (Fig. 4.5). Join PQ and draw $PM \perp OQ$. Then from the rt. angled ΔOMP, $MP = r \sin \delta\theta$, $OM = r \cos \delta\theta$.

$$\therefore \quad MQ = OQ - OM = r + \delta r - r \cos \delta\theta$$

$$= \delta r + r(1 - \cos \delta\theta) = \delta r + 2r \sin^2 \delta\theta/2.$$

If $\angle MQP = \alpha$, then $$\tan \alpha = \frac{MP}{MQ} = \frac{r \sin \delta\,\theta}{\delta r + 2r \sin^2 \delta\theta/2}$$

In the limit as $Q \to P$ (*i.e.* $\delta\theta \to 0$), the chord PQ turns about P and becomes the tangent at P and $\alpha \to \phi$.

$$\therefore \quad \tan \phi = \underset{Q \to P}{\text{Lt}} (\tan \alpha) = \underset{\delta\theta \to 0}{\text{Lt}} \frac{r \sin \delta\theta}{\delta r + 2r \sin^2 \delta\theta/2}$$

$$= \underset{\delta\theta \to 0}{\text{Lt}} \frac{r\,(\sin \delta\theta/\delta\theta)}{(\delta r/\delta\theta) + r \sin \delta\theta/2 \cdot (\sin \delta\theta/2 \div \delta\theta/2)}$$

$$= \frac{r \cdot 1}{(dr/d\theta) + r \cdot 0 \cdot 1} = r \frac{d\theta}{dr}$$

Cor. Angle of intersection of two curves. *If ϕ_1, ϕ_2 be the angles between the common radius vector and the tangents to the two curves at their point of intersection, then the angle of intersection of these curves is $\phi_1 \sim \phi_2$.*

(2) Length of the perpendicular from pole on the tangent. *If p be the perpendicular from the pole on the tangent, then*

(*i*) $\mathbf{p = r \sin \phi}$ (*ii*) $\dfrac{\mathbf{1}}{\mathbf{p^2}} = \dfrac{\mathbf{1}}{\mathbf{r^2}} + \dfrac{\mathbf{1}}{\mathbf{r^4}} \left(\dfrac{\mathbf{dr}}{\mathbf{d\theta}}\right)^{\mathbf{2}}$

From the rt. $\angle$ed ΔOTP, $p = r \sin \phi$

$$\therefore \quad \frac{1}{p^2} = \frac{1}{r^2} \operatorname{cosec}^2 \phi = \frac{1}{r^2}(1 + \cot^2 \phi)$$

$$= \frac{1}{r^2}\left[1 + \frac{1}{r^2}\left(\frac{dr}{d\theta}\right)^2\right] = \frac{1}{r^2} + \frac{1}{r^4}\left(\frac{dr}{d\theta}\right)^2 \quad \text{[by (1)}$$

(3) Polar subtangent and subnormal. Let the tangent and the normal at any point $P(r, \theta)$ of a curve meet the line through the pole perpendicular to the radius vector OP in T and N respectively (Fig. 4·6). Then OT is called the *polar subtangent* and ON the *polar subnormal*.

Let $\angle OTP = \phi$ so that $\tan \phi = r d\theta/dr$

Clearly $\angle PNO = \phi$.

$\therefore$ (*i*) **Polar subtangent**

$$= OT = r \tan \phi = r \cdot r\, d\theta/dr = \mathbf{r^2 \frac{d\theta}{dr}}$$

(*ii*) **Polar subnormal**

$$= ON = r \cot \phi = r \cdot \frac{1}{r}\frac{dr}{d\theta} = \mathbf{\frac{dr}{d\theta}}$$

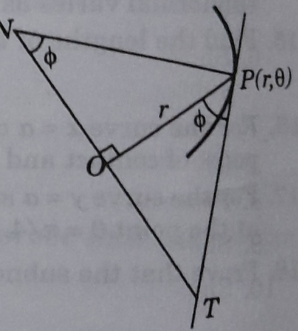


Fig. 4·6.

Example 4·39. *For the cardioid $r = a(1 - \cos \theta)$, prove that*

(*i*) $\phi = \theta/2$ (*ii*) $p = 2a \sin^3 \theta/2$.

(*iii*) *polar subtangent* $= 2a \sin^2 \dfrac{\theta}{2} \tan \dfrac{\theta}{2}$.

Sol. We have $\dfrac{dr}{d\theta} = a \sin \theta$

$$\therefore \quad \tan \phi = r \frac{d\theta}{dr} = a(1 - \cos \theta) \cdot \frac{1}{a \sin \theta}$$

$$= 2 \sin^2 \theta/2 \div 2 \sin \theta/2 \cos \theta/2 = \tan \theta/2. \text{ Thus } \phi = \theta/2 \quad ...(i)$$

Also $p = r \sin \phi = a(1 - \cos \theta) \cdot \sin \theta/2 = a \cdot 2 \sin^2 \theta/2 \cdot \sin \theta/2$

$$= 2a \sin^3 \theta/2 \quad ...(ii)$$

Polar subtangent $= r^2\, d\theta/dr = [a(1 - \cos \theta)]^2 \div a \sin \theta$

$$= 4a \sin^4 \theta/2 \div 2 \sin \theta/2 \cos \theta/2 = 2a \sin^2 \theta/2 \tan \theta/2. \quad ...(iii)$$

Example 4·40. *Find the angle of intersection of the curves $r = \sin \theta + \cos \theta$, $r = 2 \sin \theta$.*

(*Mysore, 1999*)

Sol. To find the point of intersection of the curves $r = \sin\theta + \cos\theta$ ...(i)

and $r = 2\sin\theta$, ...(ii), we eliminate r.

Then $2\sin\theta = \sin\theta + \cos\theta$ or $\tan\theta = 1$ *i.e.* $\theta = \pi/4$.

For (i), $\dfrac{dr}{d\theta} = \cos\theta - \sin\theta$

$\therefore \quad \tan\phi = r\dfrac{d\theta}{dr} = \dfrac{\sin\theta + \cos\theta}{\cos\theta - \sin\theta}$ which $\to \infty$ at $\theta = \pi/4$. Thus $\phi = \pi/2$.

For (ii), $dr/d\theta = 2\cos\theta \quad \therefore \tan\phi' = r\dfrac{d\theta}{dr} = \dfrac{2\sin\theta}{2\cos\theta} = 1$ at $\theta = \pi/4$. Thus $\phi' = \pi/4$

Hence the angle of intersection of (i) and (ii) $= \phi - \phi' = \pi/4$.

Problems 4.9

1. For a curve in Cartesian form, show that $\tan\phi = \dfrac{xy' - y}{x + yy'}$.
2. Show that in the equiangular spiral $r = ae^{\theta \cot\alpha}$, the tangent is inclined at a constant angle to the radius vector. (*Marathwada, 1990*)
3. Show that the tangent to the cardioid $r = a(1 + \cos\theta)$ at the point $\theta = \pi/3$ and $\theta = 2\pi/3$ are respectively parallel and perpendicular to the initial line. (*V.T.U., 2006*)
4. Prove that, in the parabola $2a/r = 1 - \cos\theta$,
   (i) $\phi = \pi - \theta/2$, (ii) $p = a \operatorname{cosec} \theta/2$, and (iii) polar subtangent $= 2a \operatorname{cosec}\theta$.
5. Show that the angle between the tangent at any point P and the line joining P to the origin is the same at all points of the curve
   $$\log(x^2 + y^2) = k\tan^{-1}(y/x).$$ (*A.M.I.E., 1997*)
6. Show that in the curve $r = a\theta$, the polar subnormal is constant and in the curve $r\theta = a$ the polar subtangent is constant.
7. Find the angle of intersection of the curves
   (i) $r = 2\sin\theta$, and $r = 2\cos\theta$ (*Bhopal, 1991*) (ii) $r = a(1 + \sin\theta)$ and $r = a(1 - \sin\theta.)$ (*V.T.U., 2000*)
8. Prove that the curves $r = a(1 + \cos\theta)$ and $r = b(1 - \cos\theta)$ intersect at right angles. (*V.T.U., 2006*)
9. Show that the curves $r^n = a^n \cos n\theta$ and $r^n = b^n \sin n\theta$ cut each other orthogonally. (*Bangalore, 1993*)
10. Show that the angle of intersection of the curves $r = a\log\theta$ and $r = a/\log\theta$ is $\tan^{-1}[2e/(1 - e^2)]$ (*V.T.U., 2005*)

4.8. PEDAL EQUATION

If r be the radius vector of any point on the curve and p, the length of the perpendicular from the pole on the tangent at that point, then *the relation between p and r is called the pedal equation of the curve.*

Given the cartesian or polar equation of a curve, we can derive its pedal equation. The method is explained through the following examples.

Example 4.41. *Find the pedal equation of the ellipse* $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1.$ ...(i)

Sol. Equation of the tangent at (x, y) is $\dfrac{Xx}{a^2} + \dfrac{Yy}{b^2} = 1$ ...(ii)

p, length of $\perp$ from $(0, 0)$ on (ii) $= \dfrac{-1}{\sqrt{[(x/a^2)^2 + (y/b^2)^2]}}$

or $$\frac{1}{p^2} = \frac{x^2}{a^4} + \frac{y^2}{b^4}$$ ...(iii)

$$r^2 = x^2 + y^2 \qquad \ldots(iv)$$

Also

Then from (i), $\frac{y^2}{b^2} = \frac{a^2 - r^2}{a^2 - b^2}$

Substituting the value of y^2 from (iv) in (i),

$$\frac{x^2}{a^2} = \frac{r^2 - b^2}{a^2 - b^2}.$$

Now substituting these values of x^2/a^2 and y^2/b^2 in (iii),

$$\frac{1}{p^2} = \frac{1}{a^2}\left(\frac{r^2 - b^2}{a^2 - b^2}\right) + \frac{1}{b^2}\left(\frac{a^2 - r^2}{a^2 - b^2}\right)$$

or

$$\frac{a^2 b^2}{p^2} = \frac{r^2 b^2 - b^4 + a^4 - a^2 r^2}{a^2 - b^2} = a^2 + b^2 - r^2$$

Hence $\frac{a^2 b^2}{p^2} = a^2 + b^2 - r^2$ **which is the required pedal equation.**

Example 4·42. *Find the pedal equation of the curves*

(i) $2a/r = 1 - \cos\theta$. *(ii)* $r^n = a^n \cos n\theta$ *(V.T.U., 2000)*

Sol. (i) We have
$$2a/r = 1 - \cos\theta \qquad \ldots(i)$$

Taking logs of both sides,
$$\log 2a - \log r = \log(1 - \cos\theta)$$

Differentiating both sides with respect to θ, we get
$$-\frac{1}{r}\frac{dr}{d\theta} = \frac{1}{1 - \cos\theta} \cdot \sin\theta = \cot\frac{\theta}{2}$$

$\therefore$
$$\tan\phi = r\frac{d\theta}{dr} = -\tan\theta/2 = \tan(\pi - \theta/2) \quad i.e. \quad \phi = \pi - \theta/2$$

Also
$$p = r\sin\phi = r\sin(\pi - \theta/2) \quad i.e. \quad p = r\sin\theta/2$$

or
$$p^2 = r^2\sin^2\theta/2 = r^2\left(\frac{1 - \cos\theta}{2}\right) = r^2 \cdot a/r \qquad [\text{by } (i)$$

Hence $p^2 = ar$, **which is the required pedal equation.**

(ii) From the given equation, $nr^{n-1}\frac{dr}{d\theta} = -na^n \sin n\theta$

so that
$$\tan\phi = r\,d\theta/dr = r\,\frac{nr^{n-1}}{-na^n\sin n\theta} = -\cot n\theta = \tan\left(\frac{\pi}{2} + n\theta\right)$$

i.e.
$$\phi = \pi/2 + n\theta$$

$\therefore p = r\sin\phi = r\sin\left(\frac{\pi}{2} + n\theta\right) = r\cos n\theta = r.\,(r^n/a^n) = r^{n+1}/a^n.$

Hence $p\,a^n = r^{n+1}$, **which is the required pedal equation.**

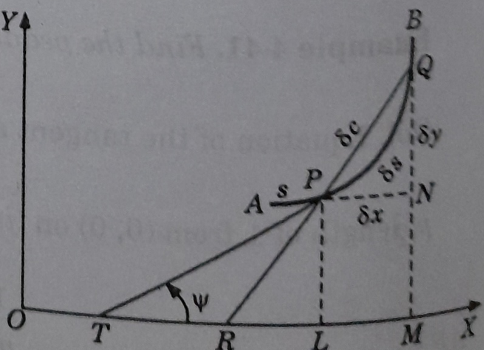
4·9. DERIVATIVE OF ARC

(1) *For the curve* $y = f(x)$, *we have*

$$\frac{ds}{dx} = \sqrt{\left[1 + \left(\frac{dy}{dx}\right)^2\right]}$$

Let $P(x, y)$, $Q(x + \delta x, y + \delta y)$ be two neighbouring points on the curve AB (Fig. 4·7). Let arc $AP = s$, arc $PQ = \delta s$ and chord $PQ = \delta c$.

Draw PL, QM ⊥s on the x-axis and $PN \perp QM$.



$\therefore$ From the rt. $\angle$ed ΔPNQ,

$$PQ^2 = PN^2 + NQ^2$$

i.e.
$$\delta c^2 = \delta x^2 + \delta y^2$$

or
$$\left(\frac{\delta c}{\delta x}\right)^2 = 1 + \left(\frac{\delta y}{\delta x}\right)^2$$

$$\therefore \quad \left(\frac{\delta s}{\delta x}\right)^2 = \left(\frac{\delta s}{\delta c} \cdot \frac{\delta c}{\delta x}\right)^2 = \left(\frac{\delta s}{\delta c}\right)^2 \left[1 + \left(\frac{\delta y}{\delta x}\right)^2\right]$$

Taking limits as $Q \to P$ (*i.e.* $\delta c \to 0$),

$$\left(\frac{ds}{dx}\right)^2 = 1 \cdot \left[1 + \left(\frac{dy}{dx}\right)^2\right] \qquad \left[\because \ \underset{\delta c \to 0}{\text{Lt}} \ \frac{\delta s}{\delta c} = 1\right.$$

If s increases with x as in Fig. 4·7, dy/dx is positive.

Thus $$\frac{ds}{dx} = \sqrt{\left[1 + \left(\frac{dy}{dx}\right)^2\right]}, \text{ taking positive sign before the radical.} \qquad ...(1)$$

Cor. 1. *If the equation of the curve is $x = f(y)$, then*

$$\frac{ds}{dy} = \frac{ds}{dx} \cdot \frac{dx}{dy} = \sqrt{\left[1 + \left(\frac{dy}{dx}\right)^2\right]} \cdot \frac{dx}{dy}$$

$$\therefore \quad \frac{\mathbf{ds}}{\mathbf{dy}} = \sqrt{\left[1 + \left(\frac{\mathbf{dx}}{\mathbf{dy}}\right)^2\right]} \qquad ...(2)$$

Cor. 2. *If the equation of the curve is in parametric form $x = f(t)$, $y = \phi(t)$, then*

$$\frac{ds}{dt} = \frac{ds}{dx} \cdot \frac{dx}{dt} = \sqrt{\left[1 + \left(\frac{dy}{dx}\right)^2\right]} \cdot \frac{dx}{dt}$$

$$= \sqrt{\left[\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dx} \cdot \frac{dx}{dt}\right)^2\right]}$$

$$\therefore \quad \frac{\mathbf{ds}}{\mathbf{dt}} = \sqrt{\left[\left(\frac{\mathbf{dx}}{\mathbf{dt}}\right)^2 + \left(\frac{\mathbf{dy}}{\mathbf{dt}}\right)^2\right]} \qquad ...(3)$$

Cor. 3. We have $\dfrac{ds}{dx} = \sqrt{\left[1 + \left(\dfrac{dy}{dx}\right)^2\right]} = \sqrt{(1 + \tan^2 \psi)} = \sec \psi$

$$\therefore \quad \cos \psi = \frac{\mathbf{dx}}{\mathbf{ds}} \qquad ...(4)$$

Also
$$\sin \psi = \tan \psi \cos \psi = \frac{dy}{dx} \cdot \frac{dx}{ds}$$

$$\therefore \quad \sin \psi = \frac{\mathbf{dy}}{\mathbf{ds}} \qquad ...(5)$$

(2) *For the curve $r = f(\theta)$, we have* $\dfrac{\mathbf{ds}}{\mathbf{d\theta}} = \sqrt{\left[\mathbf{r}^2 + \left(\dfrac{\mathbf{dr}}{\mathbf{d\theta}}\right)^2\right]}$.

Let $P(r, \theta)$, $Q(r + \delta r, \theta + \delta\theta)$ be two neighbouring points on the curve AB (Fig. 4·8). Let arc $AP = s$, arc $PQ = \delta s$ and chord $PQ = \delta c$.

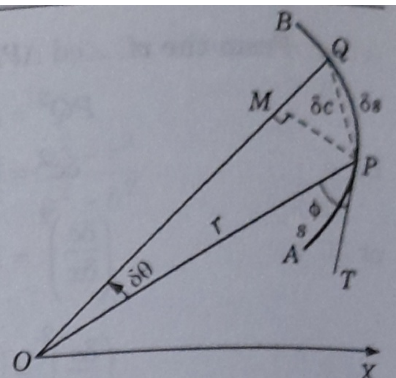


Fig. 4·8.

Draw $PM \perp OQ$, then

$$PM = r \sin \delta\theta \text{ and } MQ = OQ - OM$$
$$= r + \delta r - r \cos \delta\theta$$
$$= \delta r + 2r \sin^2 \delta\theta/2$$

From the rt. $\angle$ed $\Delta\, PMQ$,

$$PQ^2 = PM^2 + MQ^2$$

or
$$\delta c^2 = (r \sin \delta\theta)^2 + (\delta r + 2r \sin^2 \delta\theta/2)^2$$

$$\therefore \quad \left(\frac{\delta s}{\delta\theta}\right)^2 = \left(\frac{\delta s}{\delta c} \cdot \frac{\delta c}{\delta\theta}\right)^2 = \left(\frac{\delta s}{\delta c}\right)^2 \left[\left(\frac{r \sin \delta\theta}{\delta\theta}\right)^2 + \left(\frac{\delta r}{\delta\theta} + \frac{2r \sin^2 \delta\theta/2}{\delta\theta}\right)^2\right]$$

$$= \left(\frac{\delta s}{\delta c}\right)^2 \left[r^2 \left(\frac{\sin \delta\theta}{\delta\theta}\right)^2 + \left(\frac{\delta r}{\delta\theta} + r \sin \frac{\delta\theta}{2} \cdot \frac{\sin \delta\theta/2}{\delta\theta/2}\right)^2\right]$$

Taking limits as $Q \to P$

$$\left(\frac{ds}{d\theta}\right)^2 = 1^2 \cdot \left[r^2 \cdot 1^2 + \left(\frac{dr}{d\theta} + r \cdot 0 \cdot 1\right)^2\right] = r^2 + \left(\frac{dr}{d\theta}\right)^2$$

As s increases with the increase of θ, $ds/d\theta$ is positive. Thus

$$\frac{ds}{d\theta} = \sqrt{\left[r^2 + \left(\frac{dr}{d\theta}\right)^2\right]} \qquad ...(1)$$

Cor. 1. If the equation of the curve is $\theta = f(r)$, then

$$\frac{ds}{dr} = \frac{ds}{d\theta} \cdot \frac{d\theta}{dr} = \sqrt{\left[r^2 + \left(\frac{dr}{d\theta}\right)^2\right]} \cdot \frac{d\theta}{dr}$$

$$\therefore \quad \mathbf{\frac{ds}{dr} = \sqrt{\left[1 + r^2 \left(\frac{d\theta}{dr}\right)^2\right]}} \qquad ...(2)$$

Cor. 2. We have $\dfrac{ds}{dr} = \sqrt{\left[1 + \left(r \dfrac{d\theta}{dr}\right)^2\right]} = \sqrt{[1 + \tan^2 \phi]} = \sec \phi$

$$\therefore \quad \mathbf{\cos \phi = \frac{dr}{ds}} \qquad ...(3)$$

Also $\quad \sin \phi = \tan \phi \cdot \cos \phi = r \dfrac{d\theta}{dr} \cdot \dfrac{dr}{ds}$

$$\therefore \quad \mathbf{\sin \phi = r \frac{d\theta}{ds}} \qquad ...(4)$$

Problems 4·10

Prove that the pedal equation of :

1. the parabola $y^2 = 4a(x + a)$ is $p^2 = ar$.

2. the hyperbola $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$ is $a^2 b^2/p^2 = r^2 - a^2 + b^2$.

3. the astroid $x = a \cos^3 t, y = a \sin^3 t$ is $r^2 = a^2 - 3p^2$.

Find the pedal equations of the following curves :

4. $r = a(1 + \cos\theta)$

5. $r^2 = a^2 \sin^2 \theta$